



General Education: MATHEMATICS

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Solution

61. One angle of a parallelogram is 35. What are the measure of the three other angles?

A. $85^\circ, 135^\circ, 140^\circ$

B. $145^\circ, 35^\circ, 145^\circ$

C. $35^\circ, 65^\circ, 65^\circ$

D. $45^\circ, 65^\circ, 170^\circ$



Opposite angles of a parallelogram are equal and adjacent angles are supplementary.

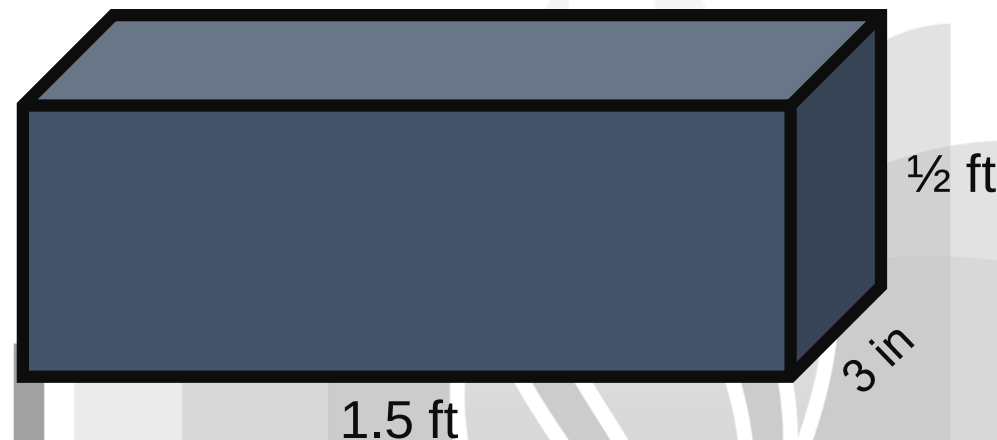
$$b = 35^\circ$$

$$a = c = 180^\circ - 35^\circ = 145^\circ$$



62. The edge of a rectangular solid have these measures:
 $1.5 \text{ ft} \times \frac{1}{2} \text{ ft} \times 3 \text{ inches}$. What is its volume in cubic inches?

- A. 225 C. 324
B. 272 D. 27



Conversion

$$1.5 \text{ ft} \times \frac{12 \text{ inches}}{1 \text{ ft}} = 18 \text{ inches}$$

$$\frac{1}{2} \text{ ft} \times \frac{12 \text{ inches}}{1 \text{ ft}} = 6 \text{ inches}$$

Solution

$$V = lwh = (18)(6)(3) = 324 \text{ in}^3$$



63. In a certain school, the ratio of boys to girls is 3 to 7. If there are 150 boys and girls in the school, how many boys are there?

A. 75

B. 90

C. 105

D. 45

Solution

$$3 + 7 = 10$$

$$150 \div 10 = 15$$

$$(3)(15) = 45$$



Solution

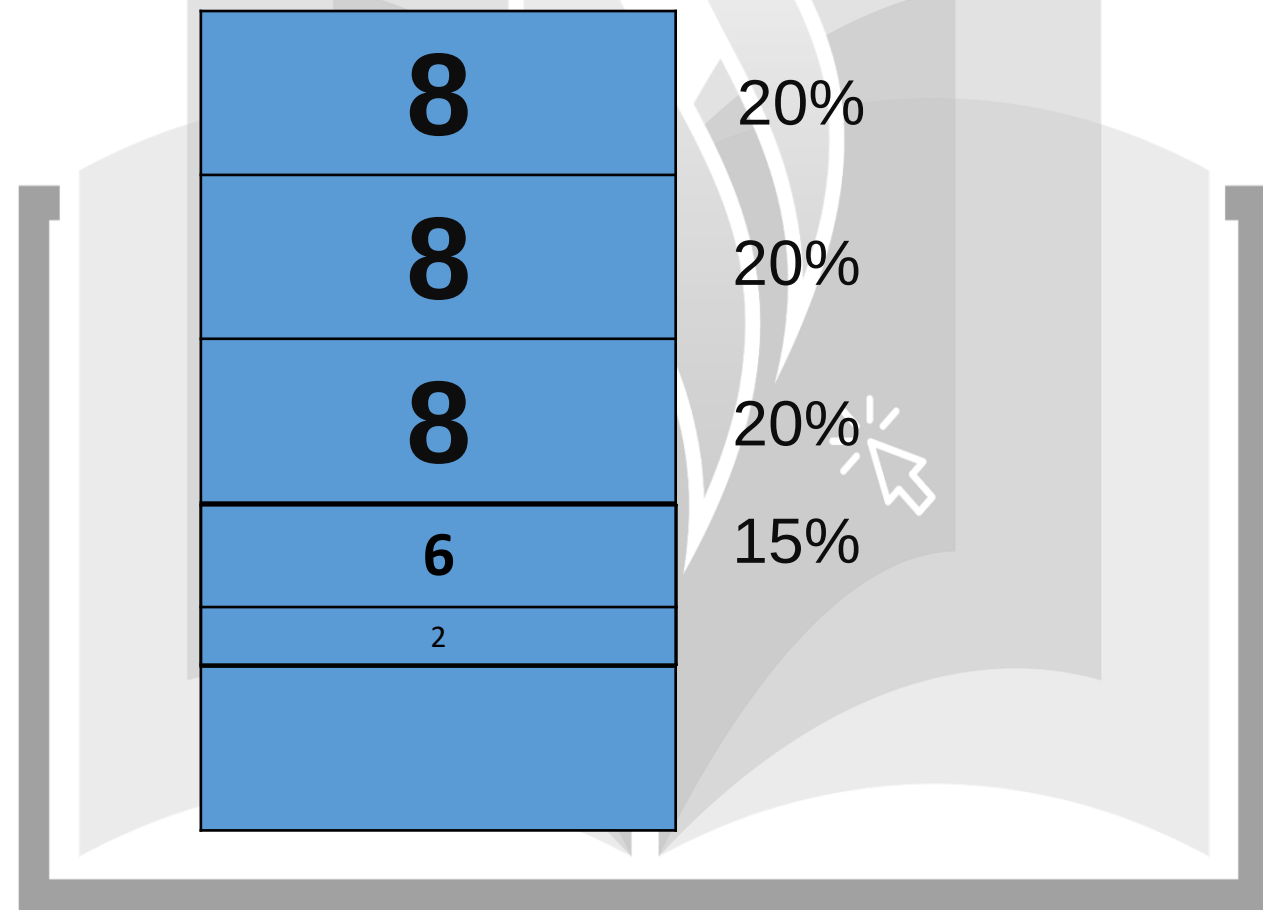
64. A water tank contains 8 liters when it is 20% full. How many liters does it contain when it is 75% full?

A. 58

C. 30

B. 15

D. 60



$$8 + 8 + 8 + 6 = 30$$



65. Which of the following has the largest value?

A. 3^9

C. 7^5

B. 9^3

D. 6^4

Solution

$$3^9 = 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 = \mathbf{19683}$$

$$9^3 = (3^2)^3 = 3^6$$

$$7^5 = 16807$$

$$6^4 = (3 \cdot 2)^4 = 3^4 \cdot 2^4$$



66. Which of these three fractions has the largest value?

A. $\frac{3}{4}$

B. $\frac{4}{15}$

C. $\frac{12}{24}$

D. $\frac{5}{9}$

Solution

Conversion from Fraction to Decimal

$$\frac{3}{4} = 0.75$$

$$\frac{4}{15} = 0.27$$

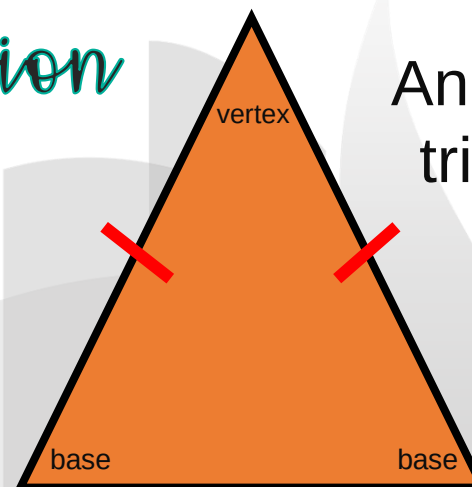
$$\frac{12}{24} = 0.5$$

$$\frac{5}{9} = 0.55$$

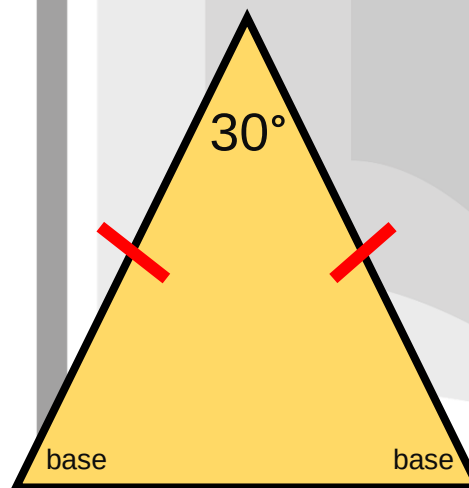


Solution

An isosceles triangle has two equal sides and angles.



Note: The sum of the angles of any triangle is 180° .



$$180^\circ - 30^\circ = 150^\circ$$

$$150^\circ \div 2 = 75^\circ$$

67. The vertex angle of an isosceles triangle is 30° . What is the measure of one of the base angles?

A. 70°

C. 150°

B. 75°

D. 60°



68. Which of these has a value different from the other three?

A. .15

C. $\frac{3}{20}$

B. $\frac{12}{75}$

D. 15%

Solution

Conversion from Fraction or Percent to Decimal

$$\frac{12}{75} = 0.16$$

$$\frac{3}{20} = 0.15$$

$$15\% = 0.15$$



69. How many degrees are between the hands of a clock at 3:40?

A. 125°

C. 130°

B. 140°

D. 150°

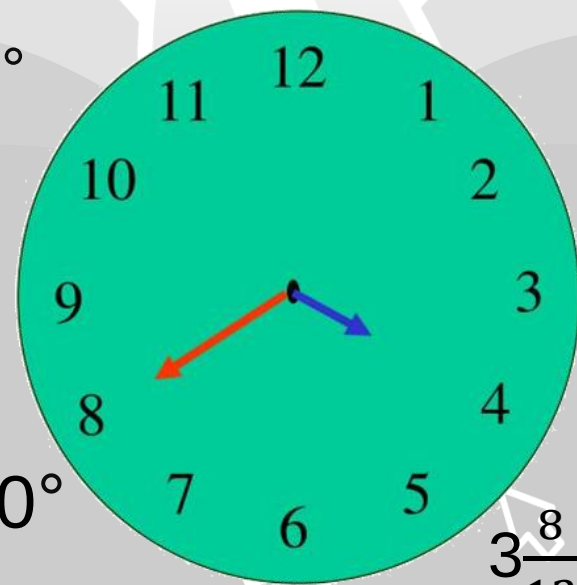
Solution

$$1 \text{ rev} = 360^\circ$$

$$8 \cdot 30^\circ = 240^\circ$$

$$3\frac{8}{12} \cdot 30^\circ = 110^\circ$$

$$240^\circ - 110^\circ = \mathbf{130^\circ}$$





Solution

70. What is the value of $12\frac{1}{6} - 2\frac{3}{8} - 7\frac{2}{3} + 19\frac{3}{4}$?

A. 22

C. 18

B. $21\frac{7}{8}$

D. 21

$$\begin{aligned} & 12\frac{1}{6} - 2\frac{3}{8} - 7\frac{2}{3} + 19\frac{3}{4} \\ &= \left(12\frac{1}{6} - 7\frac{2}{3}\right) + \left(19\frac{3}{4} - 2\frac{3}{8}\right) \\ &= 4\frac{1}{2} + 17\frac{3}{8} \\ &= 21\frac{7}{8} \end{aligned}$$



71. Which of these math sentences is NOT CORRECT?

☒ A. $1/7 + 2/3 = 3/21$

☐ B. $1/2 + 1/3 = 5/6$

☐ C. $1/6 + 1/8 = 14/48$

☐ D. $1/3 + 1/4 = 7/12$

Solution

$$1/7 + 2/3 = 17/21$$





72. What number is the next in this sequence?
11120, 11260, 11420, 11600

- A. 11760 C. 11820
B. 11720 **D. 11800**

Solution

11120

11260

11420

11600

$$11260 - 11120 = 140$$

$$11420 - 11260 = 160$$

$$11600 - 11420 = 180$$

$$11600 + 200 = \mathbf{11800}$$



73. Joseph is 10 years older than his sister. If Joseph was 25 years of age in 1983, in what year could she have been born?

- ☒ A. 1968 C. 1948
B. 1953 D. 1958

Solution

$$1983 - 25 = 1958$$

$$1958 + 10 = 1968$$



74. How many glasses of Cola each to be filled with 150 cu.cm. of the liquid can be made from 5 family size bottles of Cola each containing 1.5 liters?

A. 60

B. 45

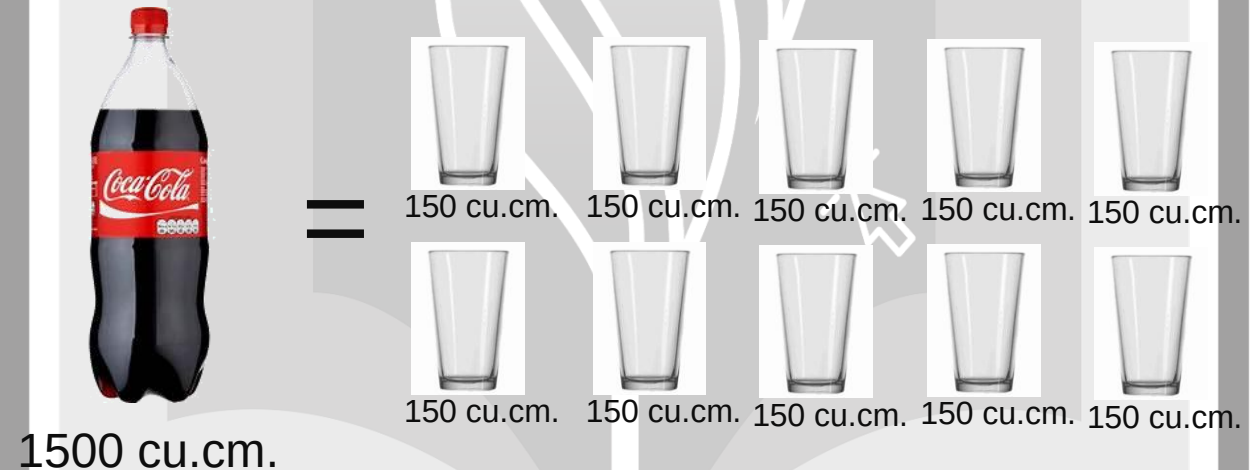
C. 50

D. 40

Solution

Conversion

$$1.5 \text{ L} \times \frac{1000 \text{ cu.cm}}{1 \text{ L}} = 1500 \text{ cu.cm.}$$



Therefore, **50 glasses** can be made from 5 family size bottles of Cola.



75. 259:37 as

A. 5:1

C. 25:4

B. 84:12

D. 63:441

Solution

$$259:37 = 7:1$$

$$5:1 = 5:1$$

$$25:4 = 25:4$$

$$84:12 = 7:1$$

$$63:441 = 1:7$$



76. What value of x will make this proportion correct?

$$1.3:x=3.25:8.00.$$

A. 20

C. 2

B. 3.5

D. 3.2

Solution

$$1.3:x = 3.25:8.00$$

$$(1.3)(8.00) = 10.4$$

$$10.4 \div 3.25 = 3.2$$



77. If the counting numbers 1 to 30 are written down, how many times does the digit 2 appear?

A. 13

C. 12

B. 11

D. 14

Solution

2	21	24	27
12	22	25	28
20	23	26	29





Mathematics

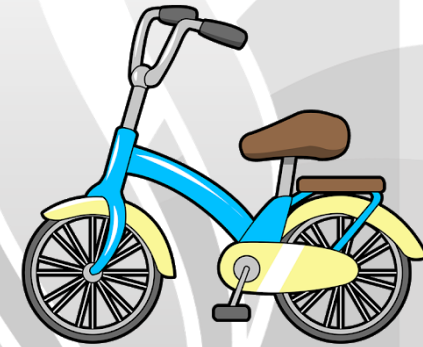
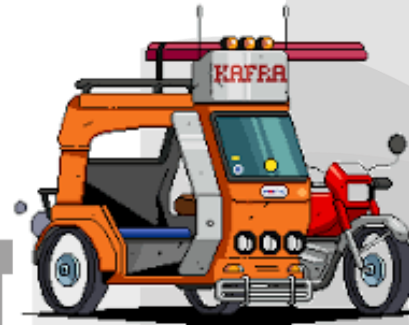
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78. In the playground for kindergarten kids, 18 children are riding tricycles or bicycles. If there are 43 wheels in all, how many tricycles are there?

A. 7
B. 8

C. 9
D. 11

Solution



10 kids
30 wheels

8 kids
16 wheels

18 kids
46 wheels

9 kids
27 wheels

9 kids
18 wheels

18 kids
45 wheels

7 kids
21 wheels

11 kids
22 wheels

18 kids
43 wheels

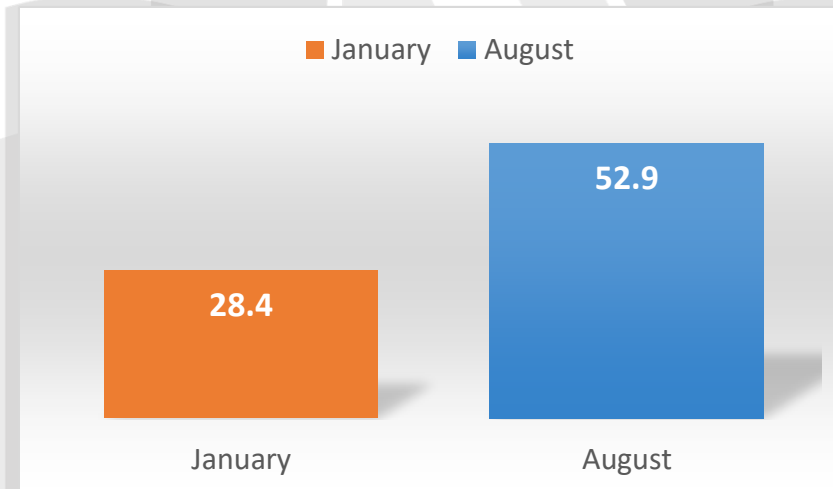


Solution

79. Between January 2008 and August 2008, the price of gasoline rose from 28.40 to 52.90 per liter. What is the percent of increase over the 8-month period?

A. 97.5% C. 46.81%

B. 86.27% D. 24.5%



$$\frac{52.90 - 28.40}{28.40} \cdot 100\% = \mathbf{86.27\%}$$



Solution

80. Ana and Beth do a job together in two hours. Working alone, Ana does the job in 5 hours. How long will it take Beth to do the job alone?

- A. 3 hours C. 3 1/3 hours
B. 2 hours D. 2 1/3 hours



Ana = 5 hrs

Beth = x hrs

Both = 2 hrs

$$\frac{1}{5} + \frac{1}{x} = \frac{1}{2}$$

$$\frac{1}{5} + \frac{1}{3} = \frac{8}{15}$$

$$\frac{1}{5} + \frac{1}{3\frac{1}{3}} = \frac{1}{2}$$

$$\frac{1}{5} + \frac{1}{2} = \frac{7}{10}$$

$$\frac{1}{5} + \frac{1}{2\frac{1}{3}} = \frac{22}{35}$$

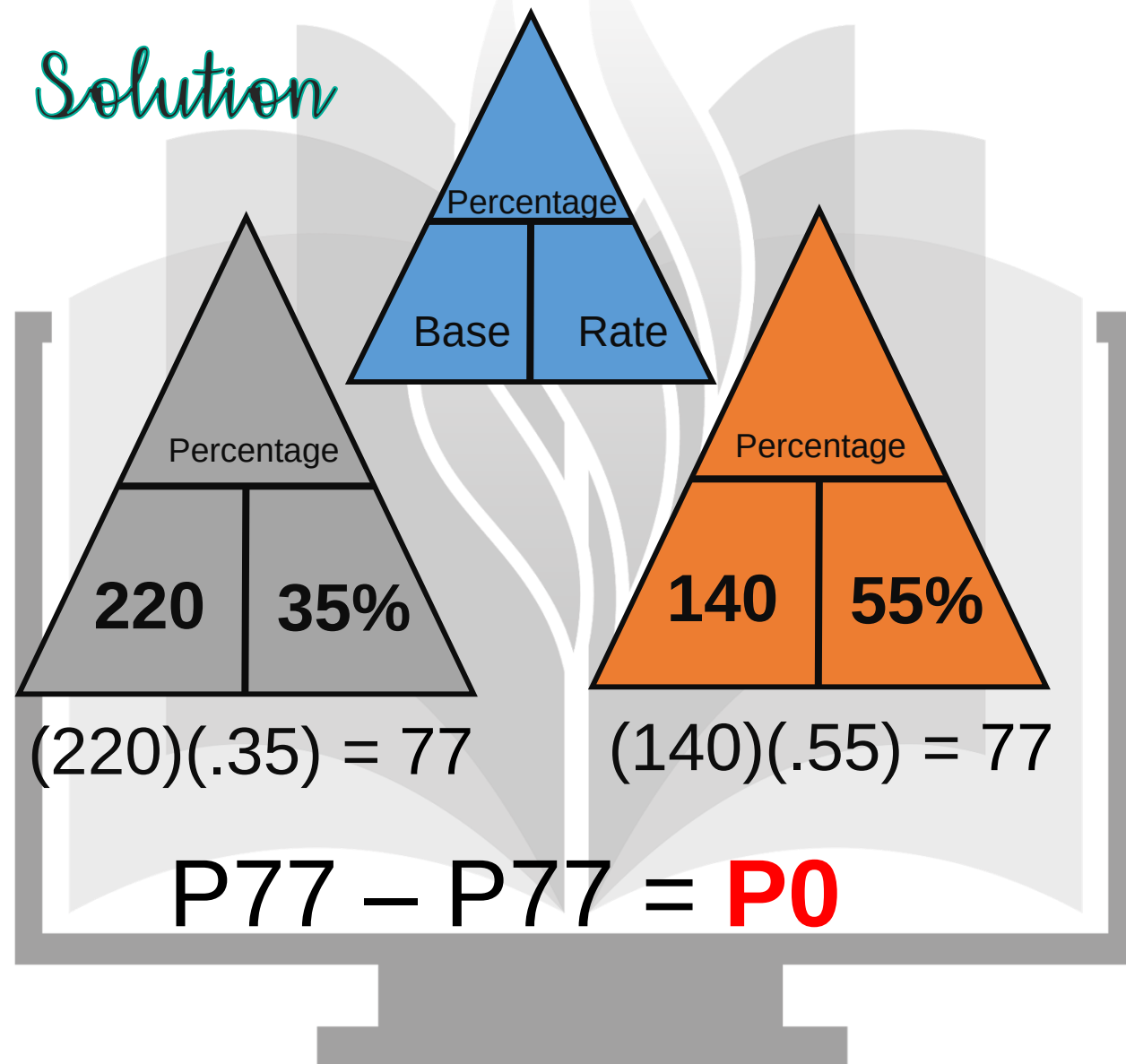


81. By how much is
35% of P220
greater than 55% of
P140?

A. P16.00 C. P480.00

B. P20.00 **D. P0.00**

Solution





82. The legs of one right triangle are 9 and 12, while those of another right triangle are 12 and 16. How much longer is the perimeter of the larger than the perimeter of the smaller triangle?

- A. 12 C. 84
B. 7 D. 14

Solution

PYTHAGOREAN THEOREM:
Given a right triangle, the square of the longest side (hypotenuse) is equal to the sum of the squares of the other legs.
 $(c^2 = a^2 + b^2)$

For the first triangle (legs 9 and 12):
 $c_1 = \sqrt{9^2 + 12^2}$
 $c_1 = \sqrt{81 + 144}$
 $c_1 = \sqrt{225}$
 $c_1 = 15$
 $P_1 = 12 + 9 + 15 = 36$

For the second triangle (legs 12 and 16):
 $c_2 = \sqrt{12^2 + 16^2}$
 $c_2 = \sqrt{144 + 256}$
 $c_2 = \sqrt{400}$
 $c_2 = 20$
 $P_2 = 16 + 12 + 20 = 48$



Mathematics

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Solution

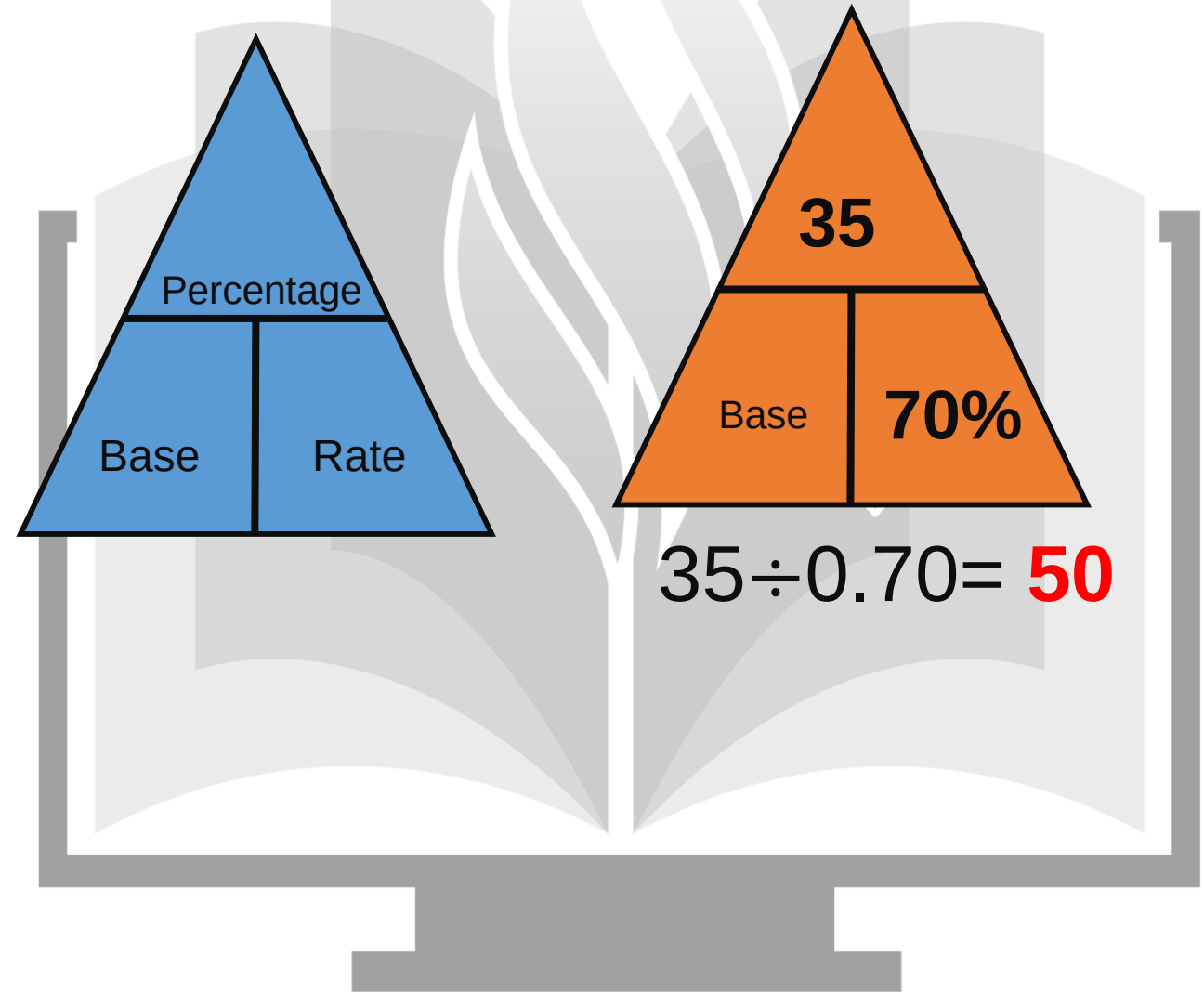
83. In a Mathematics test, 35 students passed and 30% failed. How many students took the test?

A. 60

C. 30

B. 40

D. 50





84. Each bag of rice contains 50 kg. What would be the increase in price per kilo if the cost of one bag of rice goes from P950.00 to P1,200.00?

- A. P250.00 C. P3.75
B. P7.50 **D. P5.00**

Solution

$$P1,200.00 - P950.00 = P250.00$$

$$P250.00 \div 50 = \text{P5}$$



85. How much greater is the sum of the first 20 counting numbers greater than the sum of the first 10 counting numbers?

A. 55

C. 100

B. 155

D. 110

Solution

$$S = \frac{n}{2} (a_1 + a_n)$$

$$S_{20} = \frac{20}{2} (1 + 20) = 210$$

$$S_{10} = \frac{10}{2} (1 + 10) = 55$$

$$210 - 55 = 155$$



86. Which of these numbers is a perfect square number?

A. 10217356

B. 2916488

C. 35366809

D. 262758

Solution

$$1^2 = 1$$

$$2^2 = 4$$

$$3^2 = 9$$

$$4^2 = 16$$

$$5^2 = 25$$

$$6^2 = 36$$

$$7^2 = 49$$

$$8^2 = 64$$

$$9^2 = 81$$

$$0^2 = 0$$

$$19^2 \cdot 313^2 = 35366809$$



87. If a faucet drips at the rate of 3 drops every 5 seconds, how many drops will fall in one minute?

A. 36

C. 15

B. 60

D. 100

Solution

$$3:5 = \frac{\quad}{\quad} : 60$$

$$3(60) = 180$$

$$180 \div 5 = 36$$



88. How many vertices does a rectangular solid have?

A. 6

C. 4

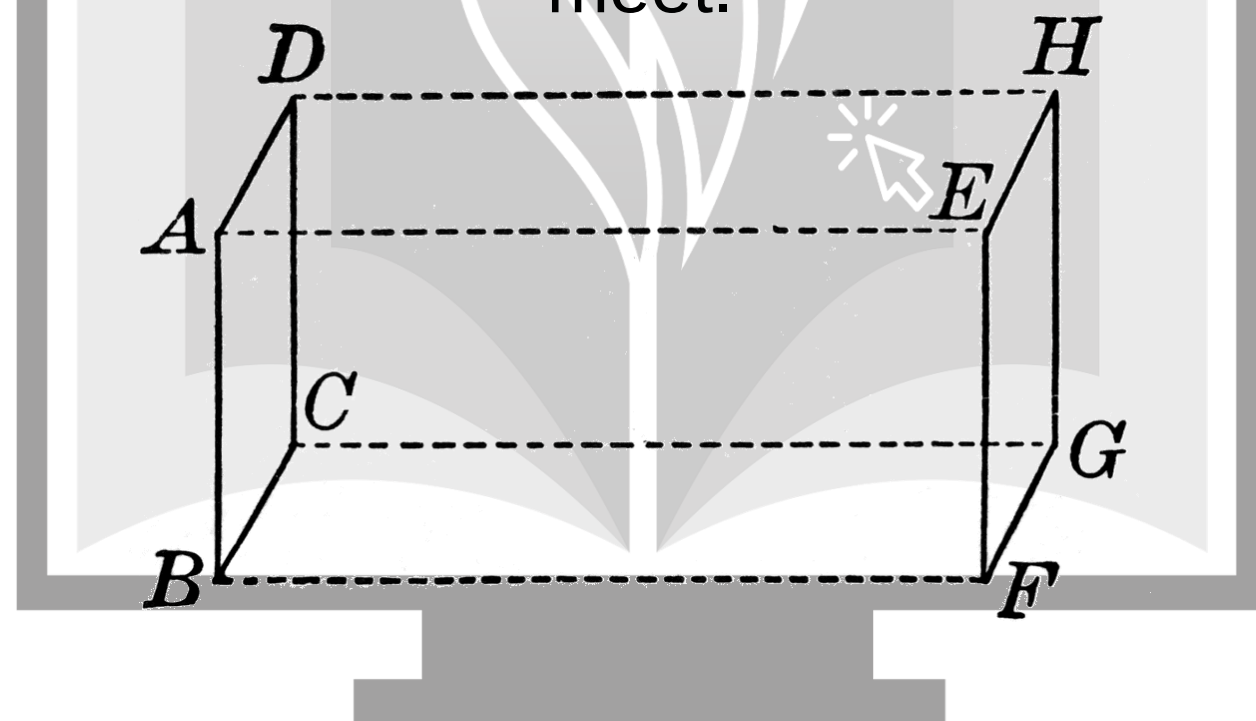
B. 12

D. 8

Solution

Vertices

Points where two or more edges meet.





89. If July 12, 2003 fell on a Tuesday, on what day will July 12, 2009 fall?

A. Monday

B. Sunday

C. Wednesday

D. Thursday

Solution

$$2009 - 2003 = 6$$

Note:

2004 and 2008 are leap years which means that an extra day is added to each year.

Hence, two extra days will be added.

$6 + 2 = 8$ days after Tuesday

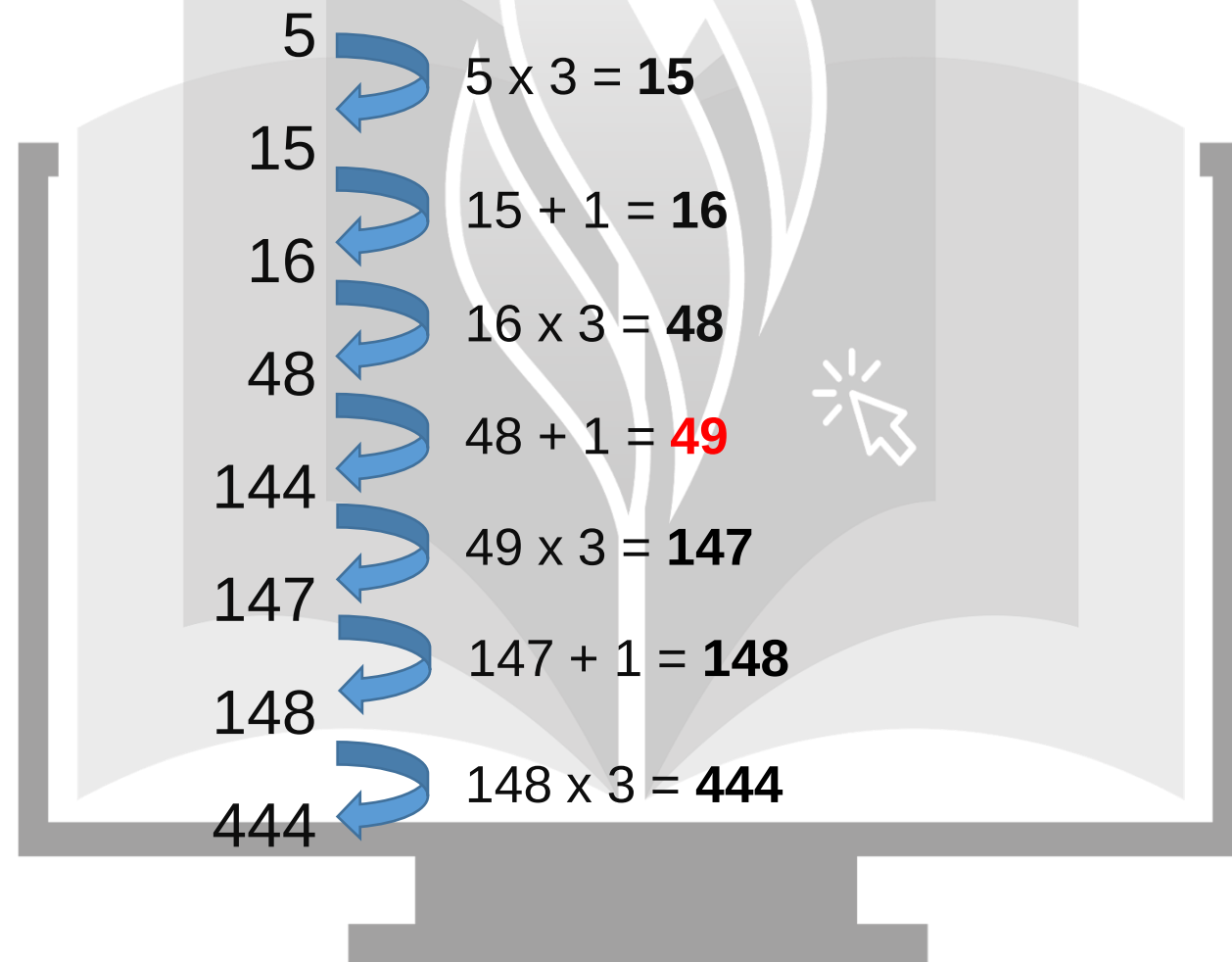


90. Which number is wrong in the sequence? What should the number be for the sequence to be correct?

5, 15, 16, 48, 144, 147, 148, 444...

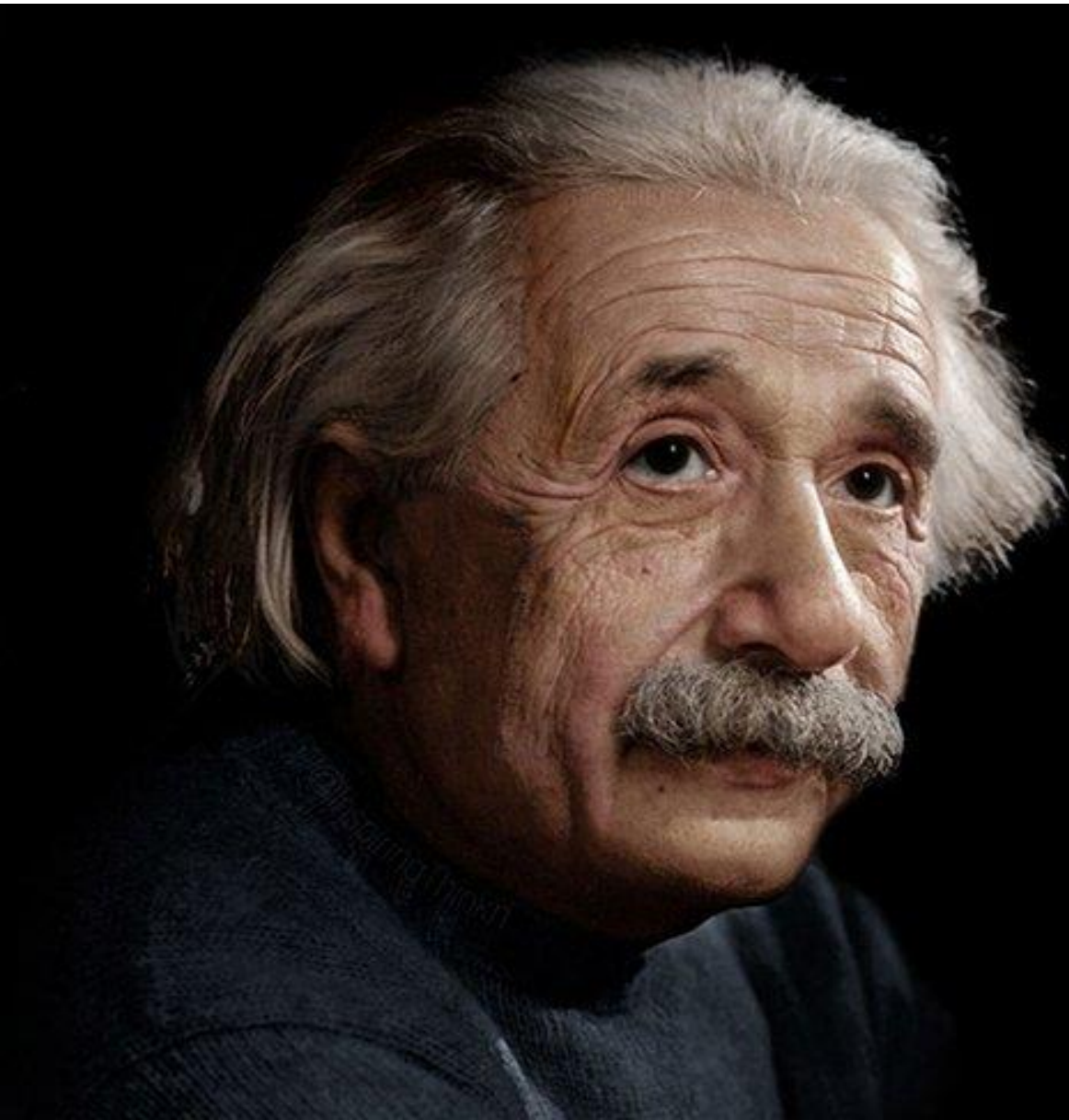
- A.** 144; it should be 49
- B. 144; it should be 132
- C. 48; it should be 18
- D. 147; it should be 145

Solution





General Education: MATHEMATICS



It's not that I'm
so smart, it's
just that I stay
with problems
longer.

Albert Einstein / @InspiringThinkn